Shreyansh Shekhar Dwivedi

Software Developer & Researcher

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PROFESSIONAL SUMMARY

B.Tech CS (Data Science) student specializing in backend engineering, agentic AI systems, and quantum-inspired cryptography. Currently Software Intern at CapConnect+ and former Research Intern at IIT Bhilai (transmon qubit ML research). Proven open-source contributor to TensorFlow (Google), Eclipse Foundation, and OWASP. Skilled in Spring Boot microservices, RAG pipelines, Docker/Kubernetes, and secure distributed architectures.

TECHNICAL SKILLS

Languages: Java (Advanced), Rust, Python, Bash
Backend & AI: Spring Boot, Spring AI, RESTful APIs, RAG, WebSockets (STOMP), DSA
DevOps & Cloud: Docker, Kubernetes, Linux, Maven, Git, CI/CD
Databases: PostgreSQL, MongoDB, ChromaDB
Security / Crypto: AES-256-GCM, SHA-256, BB84 QKD Simulation, Penetration Testing
LLM Integration: RAG Pipelines, Embedding Search, OpenAI API, Gemini API

WORK EXPERIENCE

Software Intern — CapConnect+ *Aug 2026 – Present*
- Contributing to full-stack feature development and code quality initiatives as part of the engineering team, applying Java/Spring Boot and React expertise in a production environment.

Research Intern — IIT Bhilai (Online Programs) *Jun 2026 – Jul 2026*
- Researched transmon qubit architecture and quantum coherence using machine learning; built a full ML pipeline (Random Forest, XGBoost with SHAP, Graph Neural Network) to predict materials that improve qubit coherence.

Independent Researcher — Quantum-Inspired Cryptography *Nov 2025 – Aug 2026*
- Designed QI-KDF: a 6-stage deterministic key derivation function combining BigDecimal precision seeding with microsecond temporal entropy — achieving a 2^190.5 key space, zero modulo bias, and ~500K ops/s throughput.
- Verified strict SHA-256 avalanche criterion (~50.1% flip rate) across 10K+ trials; roadmap targets SHA-3 migration, CRYSTALS-Kyber wrapping, QRNG integration, and RFC/NIST standardization submission.

Open Source Contributor — TensorFlow (Google), Eclipse Foundation, OWASP *Nov 2025 – Mar 2026*
- TensorFlow (Google): Resolved CI/CD 'run format check' pipeline failures; integrated Maven Spotless for automated style enforcement.
- Eclipse Foundation (4diac IDE): 4 merged internationalization (i18n) PRs, reviewed by maintainer Alois Zoitl.
- OWASP FinBot CTF: 3 merged PRs (#265–#267) fixing f-string interpolation bugs in audit trails, each with unit tests.

PROJECTS

ARIA — Adaptive Realtime Intelligence Agent *Personal Project | 2025 – Present*
- Built an OS-level agentic AI with LLM-controlled terminal access and a READ/WRITE/DESTRUCTIVE intent classifier that gates every command pre-execution, preventing unrecoverable system mutations.
- Spring Boot 3.x backend + Python/ChromaDB RAG sidecar for persistent memory; React/Tauri desktop UI with offline Porcupine wake-word detection; MongoDB audit trail — fully air-gapped capable.

Stack: *Java 21, Spring Boot 3.x, Python, ChromaDB, RAG, React, Tauri, MongoDB, Porcupine*

Multi-Modal RAG & AI Chat Service *Jan 2026*
- Built a RAG pipeline over private datasets using embedding-based similarity search to minimize hallucinations; engineered Spring @Configuration / @Qualifier multi-provider AI bean management (Gemini + OpenAI).

Stack: *Java 21, Spring Boot, Spring AI, OpenAI API, Gemini API, Embedding Search*

EDUCATION

B.Tech — Computer Science Engineering (Data Science) | JSS Academy of Technical Education, Noida | Expected 2028 | CGPA: 7.31

ACHIEVEMENTS

- Merged contributions into TensorFlow (Google), Eclipse Foundation, and OWASP as a first-year undergraduate — spanning three enterprise-scale open-source ecosystems.
- Authored original cryptography research (QI-KDF): 2^190.5 key space, zero modulo bias, ~500K ops/s — addressing a gap no existing RFC standard covers.
- Designed and shipped ARIA, an OS-level agentic AI with intent-aware pre-execution guardrails, cited as a working proof-of-concept in a GSoC 2026 OWASP proposal.
- Recognised with the GitHub Pull Shark badge for open-source contributions to TensorFlow Java.